CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 20: Cylinder, Cone and Sphere (Surface Area and Volume)

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Find the curved surface area and the volume of a cylinder of radius 3.5 cm and height 12 cm.

2. The curved surface area of a cylinder of height 14 cm is 88 cm². Find its radius.

3. A cone has radius 5 cm and slant height 13 cm. Find its height and its volume in terms of π.

4. Find the total surface area of a cone of radius 6 cm and height 8 cm. Take $\pi = 3.14$.

5. Find the surface area and the volume of a sphere of radius 7 cm.

6. Find the total surface area of a solid hemisphere of radius 14 cm.

7. A hollow cylinder is 35 cm long, with external radius 5 cm and internal radius 4 cm. Find the volume of material.

8. How many spherical balls of radius 1 cm can be made by melting a solid cylinder of radius 4 cm and height 6 cm?

9. A solid cone of radius 6 cm and height 24 cm is melted and recast into a sphere. Find the radius of the sphere.

10. A cylindrical vessel of radius 7 cm contains water. A solid cone of radius 3.5 cm and height 12 cm is fully immersed. Find the rise in the water level.

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Answer Key

1. CSA 264 cm²; volume 462 cm³
2. 1 cm
3. $h = 12$ cm; volume $= 100\pi$ cm³
4. $l = 10$ cm; $3.14 \times 6 \times 16 = 301.44$ cm²
5. 616 cm²; $\frac{4312}{3} \approx 1437.33$ cm³
6. $3 \times \frac{22}{7} \times 196 = 1848$ cm²
7. $\frac{22}{7} \times 9 \times 35 = 990$ cm³
8. $\frac{96\pi}{\frac{4}{3}\pi} = 72$
9. $288\pi = \frac{4}{3}\pi r^3$, so $r^3 = 216$ and $r = 6$ cm
10. $\frac{49\pi}{49\pi} = 1$ cm

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